

Empirical Lawhood

Evidence and provenance

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1 Scope and release assumption

This note maps the revised manuscript to retained documentary sources, preserves the reported results' inferential scope, and distinguishes the planned evidence release from the files used to prepare this edition.

The manuscript is drafted on the author's explicit assumption that **all referenced evidence and data will be made available in the accompanying project repository**, github.com/jokebroker/empirical_lawhood. Its availability paragraph states that release commitment. This revision does not verify that the deposit has occurred, certify repository completeness, or claim an independent experimental reproduction. No repository write or publication action was performed in preparing this bundle.

The document bundle contains the amended manuscript, its three inherited numerical figures, build materials and the documentary sources listed below. It does not contain the full native research archive. Local SHA-256 hashes authenticate the supplied documentary and figure bytes only; they do not authenticate native records merely described in those documents.

This edition revises the latest retained manuscript rather than reverting to an earlier attachment. No experiments, simulations, fitted models, calibration samples, scientific adjudications or experimental bootstrap results were added or changed. The detailed empirical sections and figure captions are preserved. The circuit history-depth account remains transcribed from the existing technical companion. The added philosophical connections, discovery framing and agency objectives are manuscript revisions, not additional experimental findings.

2 Study-to-source register

The reference numbers here match the main manuscript. Exact original filenames are retained under sources/; historical version labels have not been relabelled as new technical evidence. Several sources report overlapping populations and are not independent replications.

Main reference	Documentary source	Use
[18]	<i>Empirical Lawhood: Methods and Evidence</i> , original technical companion, 1 October	E1 circuits; E2 forecast-mediated reuse; E3 preparation and diagnostics; E4 reactor; E5 wider systems.
[19]	<i>Geometry Without Commitment</i> , 29 August	Original feasibility scan and selected-event conditional futures.
[20]	<i>Baseline-anchored, pulse-timed recursive closure</i> , 20 September	Only §§1–2 provide completed transient and baseline results. Later experimental proposals are not counted as results.
[21]	<i>Empirical Lawhood</i> , archived matrix-response account	§5 provides nonideal algebra, passive response and realised-future tangent evidence.
[22]	<i>Empirical Lawhood and Epistemic Control</i> , 6 September synthesis	§§3–4 provide matrix and cross-system evidence; historical plans remain distinct from completed studies.

Exact documentary filenames:

```
sources/Empirical_Lawhood_Methods_and_Evidence_v0.9.1.pdf
sources/geometry-without-commitment-thread-and-literature-dossier.md
sources/CC1_Baseline_Anchored_Recursive_Closure_Proposal_v2.md
sources/Empirical_Lawhood_Manuscript_v0.2(1).pdf
sources/Empirical_Lawhood_and_Epistemic_Control_White_Paper_v0.2_named.pdf
```

Two additional preserved baselines are `sources/Empirical_Lawhood_Manuscript_v0.9.4.pdf` and `sources/Evidence_Manifest_v0.9.4.pdf`. The latter contains reported native record identities and historical documentary limitations. It remains a source register, not a substitute for native records. The current manuscript’s empirical values were checked against the supplied drafts and these retained accounts, not recomputed from the full research data.

3 Empirical populations and retained distinctions

3.1 Circuit information challenge — manuscript §4.1

The analytical illustration uses equal capacitances, time constants 1 and 4, and initial branch voltages $(1, 1)V_0$ and $(0.5, 1.5)V_0$. The maximum mean-response separation is approximately $0.11812V_0$; the identical-input deterministic-prediction lower bound is $0.05906V_0$. These are analytical quantities, not sampled trajectories.

The numerical challenge retains **36 independent seed blocks** and **2,319 confirmed nominations**: 795 nonadverse, 625 response-adverse only and 899 also decision-adverse. All eighteen family/size/endpoint comparisons have witnesses in all twelve applicable seed units, with reported simultaneous one-sided lower bound 0.6123. The comparisons reuse seed units. The independent population is not eighteen new populations of twelve blocks.

The complete entered history-depth account from [18, E1.4] is:

Retained depth	Entered pairs	Response-adverse	Decision-adverse
0	864	697	432
1	567	488	279
2	432	293	144
4	188	46	44
8	144	0	0
9–13	124	0	0
16, 31, 35	Untargetable	Not observed	Not observed
Total entered	2,319	1,524	899

Observation matching in the numerical challenge is tolerance-defined. The documentary record does not establish a quantisation or observational-noise channel that makes near-equal real-valued inputs literally identical to an unrestricted predictor. Therefore the analytical all-predictor bound is not assigned to the numerical challenge. A task boundary crossed by the tested action does not imply that no common satisfactory alternative exists. Targeted witness counts are not ordinary-operation prevalence. Untargetable depths are not successful challenges or evidence that every longer history works.

3.2 Matrix geometry, algebra and response — manuscript §§4.2–4.4

The structural scan retains **9,126 valid trajectories**, of which 9,125 had neither triplet admitted and one had only the Y triplet admitted. The required recurring regimes did not qualify, and dependent partition-prediction and nested-control tests did not execute. The selected-event study retained seven cohorts of 64 conditional futures; six structurally admitted checkpoints supplied the reported **0/384** rolling-target-first

outcomes. The full 448 branches resolved, but the first-passage result is not an eventual-return probability or a new population of discovered events. [19]

The approximate-factor analysis remains a separate selected nonideal excursion, not asserted to be the isolated event above. It retains low nontrivial eigenvalues 0.1404, 0.1457 and 0.2263, a faster band beginning near 8.34, maximum principal angle 0.740° , associative leakage 2.92×10^{-4} , and the scalar diagnostic's 0.30 crossing. This is approximate factor persistence, not an exactly conserved subsystem or physical logical qubit. [21, §5.1; 22, §§4.2–4.3]

Directional results retain 64 histories and the 32-tick horizon, with median history-mean normalised losses approximately 0.00024 aggregate, 0.420 slow-direction and 3.714 worst-direction. Sixty-three histories exceed the no-change reference in mean worst-direction loss. Because normalisation uses actual increments, resolution floors and absolute task-scale errors are needed before inferring downstream task failure. Supplied passive double-commutator dynamics are not a newly discovered endogenous law. [18, E5.7; 22, §4.4]

The coupled-tangent comparison remains 6,400 nested comparisons across 32 simulation preparations. Its use of realised future paths makes it explanatory rather than a use-time forecast. The transient surrogate retains prescribed stiffness $(88/9)^2$, 93.0% description-change and 90.1% induced-response-change MSE reductions, 41.5% higher MSE than endpoint maps on their represented menu, and a baseline-to-transient squared response-error contribution ratio of approximately 781. All ten observed support failures remain incorrectly forecast inside support. Retained-data holdouts are not new independent confirmation, and no prior negative is reclassified. [20, §§1–2; 21, §5.2]

3.3 Forecast-mediated reuse — manuscript §5.1

The test retains 32 calibration and 64 fresh evaluation episodes. Preparations are assigned, not selected by a qualified applicability forecast. Downstream response coefficients remain fixed while uncertainty is calibrated for the composition. Predictions and actions are sealed before preparation; the later observed prepared state can cancel but not revise the prediction or replace the action. [18, E2]

The reported result is **61/64 joint-task successes**, one-sided 95% lower bound 0.8833, **0/64 false-admission episodes**, one-sided 95% upper bound 0.04573, 125 successful delivered assays, three pre-preparation declines and no later cancellations. All 64 episodes remain in denominators. These are matched opposite-force branch assays, not single-trajectory endpoint steering.

Prediction improves by 27.55% relative to the cached comparator and, descriptively, by 6.47% relative to direct prediction. The joint-task comparison remains 60 versus 61 successes, paired one-sided $p = 0.5$, and two changed choices among 128. Task superiority is not inferred. Simulator-available features and matched branches are not demonstrated physical sensor or intervention access.

3.4 Preparation-mediated applicability — manuscript §5.2

The prior preparation and entry-readiness negatives remain negative. The successful population is separate: eight qualification episodes are excluded from the 32 confirmation episodes, with 256 request pairs per confirmation episode, totalling 8,192 nested pairs. One selected bath recipe is shared; independent pre-observation perturbations have amplitude scale 0.002 and variance scale 4×10^{-6} . Full-strength future noise is used for response tests. Independence is conditional on that recipe and regime, not replication across different recipes or solvers. [18, E3]

The intact package—including coefficients, support, uncertainty, action set and refusal rules—is unchanged. Assay choices use the observed prepared state. H/N/P qualified successful-pair counts remain **0/7,965/0**. Primary episode validity is **0/32, 32/32, 0/32**, respectively. The paired episode-validity test has exact one-sided $p = 2.3283 \times 10^{-10}$. The one-sided 95% upper bound for zero false-admission episodes among 32 is approximately 8.94% under the stated iid model; it is not a request-level safety bound.

Bypassing only support yields H/N/P counts **7,916/7,965/7,952**, giving N minus H **49 pairs or 0.598 percentage points**. Outcome-visible action search finds suitable actions for all 8,192 pairs in all three arms.

These post hoc results diagnose the primary effect as chiefly qualification under the fixed package, not creation of task opportunity; they do not authorise unsupported use or retrospectively expand support.

Both-view support retains **31/32 episodes and 7,719 pairs**; the conservative discrepancy envelope retains **27/32 and 6,720 pairs**. Exclusions count as zero and all confirmation episodes remain in denominators. The construction establishes a fixed preparation recipe, not a calibrated law predicting preparation-dependent applicability. It has not been joined with forecast-mediated reuse as one integrated controller, and it has not been established as necessary or cheaper than independently qualifying wider support.

3.5 Wider systems — manuscript §6

Physical cortical cells, optical held-out blocks, battery cells, divertor discharges and simulation preparations retain their own acquisition hierarchies. Multiple response requests are not independent experimental replications. The manuscript preserves useful local interventions, failed voltage-error qualification, unresolved delivered-flow accounting, the reduced battery description's thermal failure, and the failed MAST-TORAX joined transfer.

Reactor qualification remains 47 of 72 context/action/horizon coordinates, with five of twelve context-horizon pairs entering prospective use. The reported 636 successful deliveries and four thermal-guard declines are among 640 planned uses across 64 episodes. The separate counterfactual audit rejected the service-gain claims while recording no false admissions for the conservative policy. Neither absent online physical resolution nor comparator service advantage is erased by simulator success. [18, E4–E5; 22, §3]

4 Historical verification issues and the repository assumption

Making records available is different from showing that every historical summary is correct. The revised manuscript therefore does not invent missing arithmetic, reassign confidence levels or claim that the publication assumption resolves a scientific inconsistency. The following issues are preserved for traceable reconciliation with the native repository records.

4.1 Nominal and final intervals in the first worked forecast episode

The retained technical companion [18, E2.8] reports the nominal longitudinal interval

$$[-0.0634282070, -0.0588729828]$$

and the realised longitudinal range approximately -0.0637118 to -0.0635472 . That displayed range is below the nominal lower endpoint. The account separately mentions an outward numerical allowance and a final expanded, domain-intersected admission box, but does not supply enough operands to reconstruct that final box unambiguously.

The revision preserves the distinction between the reported task outcomes and verification of this particular final interval. It does not infer that the nominal band was the executed admission set, assert that the executed set covered these values, add an allowance or replace the example with a more favourable one. The main manuscript cites this issue in its qualification paragraph but does not use the example as an additional positive result.

Under the planned repository release, the relevant objects are the retained point prediction, calibration score and width, deterministic allowances, domain intersection, sealed final admission set and measured outcomes for the same episode. Availability must permit reconciliation of those quantities; it does not itself establish their agreement.

4.2 Bootstrap reporting

The historical whole-episode bootstrap ranges, in percentage points, are 96.84–97.60 for primary gain, 87.93–97.55 for both-view support, 69.74–94.04 for the conservative envelope, and 0.452–0.757 for the

support-bypass N-minus-H difference. The accounts describe 20,000 resamples but do not specify a confidence level and complete construction. [18, E3.14; preserved historical evidence manifest]

The main paper therefore retains counts and explicitly specified exact one-sided bounds without labelling these bootstrap ranges 95% intervals. No resampling was performed for this revision. The repository's original analysis parameters and seeds, rather than a guessed convention, determine their interpretation.

4.3 Acquisition and independence

The retained sources also flag details needed to reconstruct training populations, selection history, the circuit generator's sign and time origin, physical cohort preprocessing, preparation schedules and the complete reactor atlas. The repository release is assumed to supply the referenced evidence and specifications. This edition does not manufacture any missing record or claim to have verified those details independently.

The internal retrospective audit shared the implementing assistant and paired-measurement code. Its authenticated replay and consistency checks are not independent physical implementation, independent-solver reproduction or external peer review. The same distinction applies after publication.

5 Research-process account and added references

5.1 Application, co-development and evidence

The revised manuscript treats the matrix programme as both an investigation of a particular medium and a human-directed application and development of the EL framework. The author supplied the account of an evolving metatheory organising questions while experimental results redirected subsequent inquiry. This is an authorial research-process account, not a new audit of the full version history, a causal estimate of the framework's contribution, or a comparative search-efficiency result.

The published sequence in *Geometry Without Commitment* supports a concrete example: failure of the original feasibility requirement was retained while the isolated structural event became a separate selected-event investigation. The algebraic and transient studies provide further documented distinctions between a detector and its object, and between an intervention's increment and its uncertain absolute baseline. These records justify describing the questions and results. They do not establish that the final framework was specified unchanged before those studies or that an autonomous selector generated the sequence.

The detailed empirical sections, captions, populations, outcome counts and inferential qualifications remain unchanged from the immediate manuscript baseline. Substrate-specific results are retained as scientific outputs. No novelty or priority judgement is upgraded by the new framing, and none of the separate constructions is joined retrospectively into an integrated controller.

5.2 Wheeler, Engelbart and meaningful human control

The main manuscript adds four references while retaining the earlier numbering:

- [25] Wheeler, *Law without Law* (1983). The account uses the discussion of chosen apparatus and registered phenomena, pp. 182–187, and the self-excited circuit and law-building conjecture, pp. 209–210. The circuit is treated as a stronger cosmological conjecture, not as an ordinary feedback diagram. No Wheeler illustration is reproduced in the manuscript.
- [26] Wheeler, *Information, Physics, Quantum: The Search for Links* (1989; related 1990 volume contribution). The consulted primary-text reprint explicitly identifies the 1989 Tokyo proceedings, pp. 354–368. Its discussions of *it from bit* and question choice provide the basis for the new subsection. The reference distinguishes this source from the related contribution listed in the 1990 Zurek volume rather than attaching the Tokyo page range to that volume.
- [27] Engelbart, *Augmenting Human Intellect* (1962). The original report supports the brief connection to improving the joint human–tool–method system. This is a relevant antecedent, not a claim that it originally motivated EL's development.

- [28] Santoni de Sio and van den Hoven (2018). Their account supports the distinction between meaningful control and mere intervention. The manuscript’s agency aims are not reported as experimental tests of that account.

The primary texts were consulted on 4 October 2026. Source locations and the specific use of each are recorded in `reference_verification.md`. Third-party papers and font files have not been added to the bundle.

5.3 Agency, autonomy and unmeasured benefits

Scientific agency is stated as a purpose: people and communities should be able to originate, understand, choose and redirect inquiry while using automated capabilities. The cockpit analogy concerns effective direction and intelligibility, not continuous manual execution or a ceremonial approval requirement. Human-directed operation is a legitimate intended design; delegated autonomous components are compatible with it.

The current evidence does not establish general accessibility, human-interface effectiveness, agency preservation, comparative computational efficiency or beneficial societal outcomes. The manuscript neither claims these results nor uses them to increase the strength of the empirical findings. Exact hardware, elapsed-development, compute, token and expenditure budgets, and associated headline comparisons, are deliberately omitted. Their later substantiation is not required to state the motivation.

Kant’s warrant question, Wheeler’s participatory conjecture and Hartle’s physical account retain distinct roles. The paper does not derive agency rights, social benefit or human authority from a physical response law. The distinction between epistemic warrant and authority to act remains explicit.

6 Editorial continuity

The early empirical overview and Neptune motivation are retained. So are the distinction among physical, informational, evidential and task-related conditions; the separation of relations, packages and tasks; and the difference between investigating a prerequisite, supplying it and qualifying a composition. The revision does not restore the earlier vague recognition formula or the argument contrasting the inquiry with an explanation of existence.

The new material is distributed across the abstract, introduction, philosophical foundations, discovery-method account, synthesis and conclusion. It is not appended as a separate manifesto. The three numerical figures and detailed experimental narratives remain the empirical centre of the paper.

7 Figures, build and identifiers

All three numerical figures are inherited byte-for-byte from the immediate manuscript bundle, which preserves the earlier v0.30 assets. No numerical marks, trajectories, uncertainty bands or values were recreated. Figure captions and the three main-text tables are unchanged in this edition. The inherited images are raster figure assets with PDF wrappers. Original figure provenance is retained in `assets/inherited_figure_provenance.json`.

The displayed title is *Empirical Lawhood*. The version is stored in filenames, source metadata and PDF metadata, not the title or running headers. The author remains Gareth Seneque, ORCID 0009-0003-3046-6899. The project/software DOI 10.5281/zenodo.23083988 is not assigned as the manuscript DOI.

`build.sh` rebuilds both PDFs with Pandoc, XeLaTeX and installed fonts. Font files are not bundled. The source inventory and bundle manifest record file hashes; the validation report concerns documents and retained numerical anchors, not experimental validation.